1 · PROBLEM AND WHO CARES

Your store knows the shopper. It knows nothing about the pet.

- Pet-store product pages are free text. "Chicken-free" is a claim, "animal fat" hides its source, and a fish-shaped toy is not fish food.
- Merchants hand-tag thousands of SKUs, or skip it. Parents of allergic pets read every label themselves.
- Nothing in the store can say, with evidence, which products suit *this* pet.

Who cares

- **Merchants** on Shopify who want pet-aware discovery without a tagging project.
- **Shoppers** who saved a pet once and expect the store to remember it.
- **Pets** with allergies, life stages and sizes a store must never guess.

Classification is a closed question, not a writing task.

- **Typed decision models** (TypeSafe Jev, open-weight Laya) answer "does this evidence support X?" with a calibrated probability in one forward pass.
- They **never generate text**. Nothing to parse, nothing invented, uncertainty you can threshold per field.
- Jev: US$0.042 per million input tokens. Laya: Apache 2.0 weights, a self-hosting path.

Why now

Shopify's agentic commerce (Global Catalog, agent profiles) makes structured, evidence-backed product attributes the currency. Laya shipped in September 2026 with free hosting through October.

Why us

Petsona already owns the pet profile inside the merchant's store, live in production, with the Finder that consumes these attributes.

Evidence in, verdicts out, merchant in control.

Product
text, variants, images, manufacturer page, label OCR

→

Evidence spans
every sentence gets an ID

→

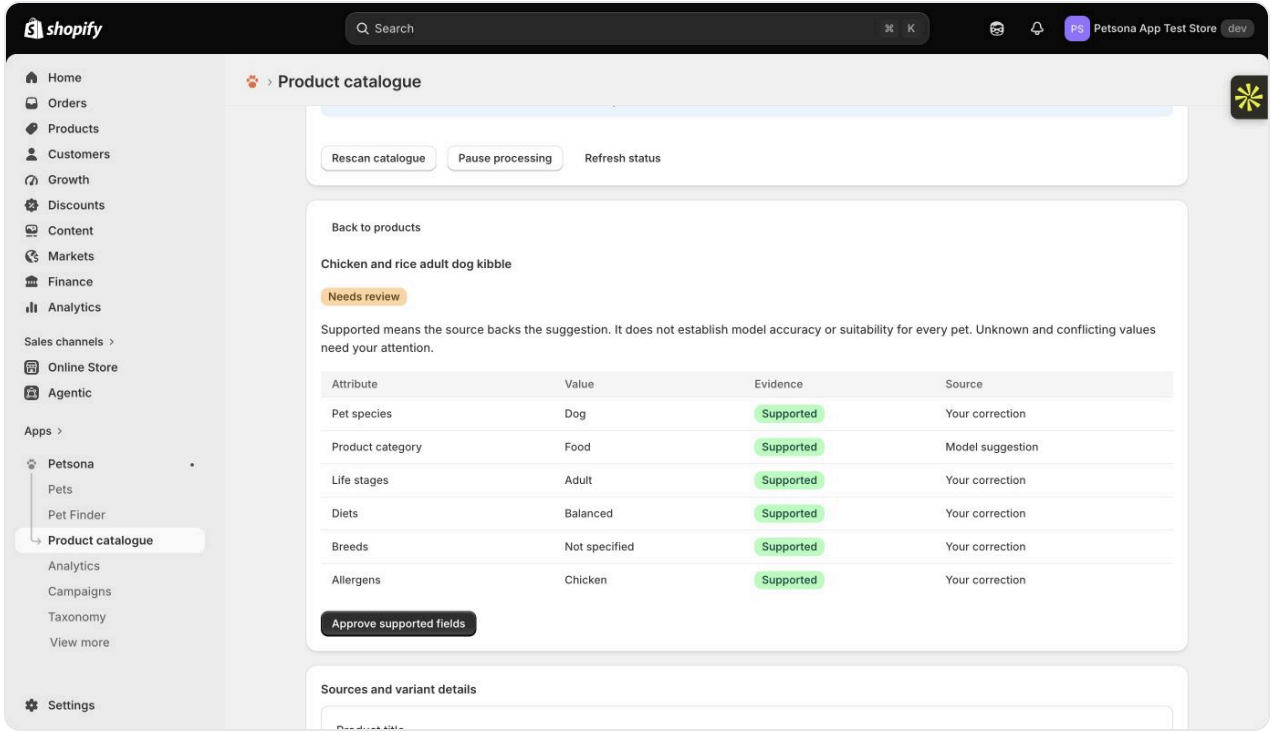
Typed verdicts
supported · contradicted · unknown · conflicting

→

Merchant review
approve, correct, lock

→

Pet Picks
only eligible products for this pet



- Review table shows the value, its evidence status and the source sentence.
- "Not suitable for kittens" beside "adult cats" is surfaced as a **conflict**, never resolved silently.
- Allergen absence is published only when the merchant attests it.
- `npm run demo:all` reproduces the pipeline in under a minute. Set a key for live verdicts.

Deterministic where it can be, typed where it must be, generative only to re-order.

Stack

- TypeScript monorepo: Shopify app (React Router), Cloud Run asia-southeast1, Neon Postgres, Cloud Tasks, daily reconciliation.
- Evidence: product text, verified manufacturer pages (robots.txt, GTIN or SKU match), Cloud Vision OCR of the merchant's own label photos.
- Decisions: one atomic four-way question per value, up to 20 per request, Jev 1.13.0 or Laya via Vercel AI Gateway, token windows with the pinned Laya tokenizer.

Guardrails

- Pinned model identity and full probability distributions checked on every response.
- Spend ledger in micro-dollars, rate-limit admission, idempotent source and plan hashes.
- Missing text is never absence. Merchant overrides lock across refreshes.
- Pet Picks filters deterministically first. A Groq-hosted LLM may only re-order at most 20 already eligible products.

115 tests in the submission repo; 1,500+ unit tests and 384 browser tests in the main repo.

Merchants pay for a store that knows the pet.

Who pays

Merchants, through Petsona's recurring plans: Free, Growth US$29, Scale US$79 per month. Catalogue intelligence sits in the paid tiers.

Unit economics

A 10,000-product catalogue at 20,000 input tokens each: about **US$8.40** per full pass on Jev. OCR: US$1.50 per 1,000 images after the free tier. Laya self-hosted: fixed infrastructure, no per-call fee.

How we reach them

- Shopify App Store listing (pet-profile app already live in production).
- Outbound to pet retailers: cold-email campaign running since October 2026.
- Theme plus app bundle for stores that want the shopper experience out of the box.
- Partner case studies from the first merchants.

Measure before we promise.

Next four weeks

- Labelled benchmark: 400 product families, 120 pet-preference cases, two annotators with adjudication, brand-separated splits.
- Per-field acceptance thresholds from measured calibration.
- Self-hosted Laya on Cloud Run before the free alias ends on 31 October.
- OCR switched on for pilot stores.

Then

Multilingual labels, variant-level sizing tables, more verified manufacturer profiles.

Risks we are naming

- Accuracy is unmeasured today: zero independently reviewed labels.
- Provider promotion ends; paid or self-hosted path must be chosen.
- Evidence status is not medical advice. Allergen absence stays a merchant attestation.
- Manufacturer pages vary. One development store so far.